

Problem 293: Autokey

Difficulty: Medium

Author: Michael Warner, Denver, Colorado, United States

Originally Published: Code Quest 2026

Problem Background

French diplomat Blaise de Vigenère created a powerful encryption cipher in 1586, the world's first "polyalphabetic substitution" cipher. Rather than replacing each letter in the original message with a single consistent ciphertext letter, each letter would be encrypted using a different cipher alphabet determined using a keyword. The cipher remained unbroken for nearly three centuries, making it one of the most resilient ciphers throughout history.

Even with its inherent strength, users of the cipher were in search of ways to strengthen it further. One such method was to use the message to encrypt itself.

Problem Description

The basic Vigenère Cipher uses a keyword to determine the cipher alphabet used to encrypt each letter in the plaintext, as shown in the table on the next page. The first letter in the keyword is used to determine the alphabet used to encrypt the first letter in the plaintext, the second keyword letter encrypts the second plaintext letter, and so on.

The "Autokey" variation of the Vigenère Cipher starts in the same manner; a keyword is used to encrypt the first few letters of the plaintext message. Usually with the Vigenère cipher, once you reach the end of the keyword, you'd start over with the first keyword letter again. With Autokey, you start using the letters in the plaintext message as the keyword instead. For example, to encrypt the message "autokey" using the keyword "CODE", you'd do the following:

Plaintext	a	u	t	o	k	e	y
Keyword	C	O	D	E	A	U	T
Ciphertext	C	I	W	S	K	Y	R

As you can see, the keyword "CODE" is used to encrypt the first four letters, but the remaining three are encrypted using "AUT" – this comes from the first three letters of the original plaintext message. In this way, we avoid repeating the keyword, and can largely avoid one of the larger vulnerabilities of the cipher (while, unfortunately, introducing new ones... but perhaps that could be a future problem).

For this problem, you'll need to encrypt messages using the Autokey cipher, given the original message and a starting keyword. Do not encrypt any spaces or punctuation, and don't include them in the encrypted text; this helps to make the cipher slightly more difficult to break.

The following table shows the arrangement of cipher alphabets; each plaintext letter is encrypted by replacing it with the letter in its column and the row for the corresponding keyword letter.

Plaintext → Keyword ↓	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Sample Input

The first line of your program's input, **received from the standard input channel**, will contain a positive integer representing the number of test cases. Each test case will include two lines of text:

- A line containing the keyword to use to start the encryption process, in uppercase letters

- A line containing the plaintext message to be encrypted, containing lowercase letters and possibly spaces and/or punctuation.

2

CODE

autokey cipher

SAMPLE

hello, world

Sample Output

For each test case, your program must print the encrypted message in all uppercase letters. Omit any characters that are not letters from the encrypted message.

CIWSKYRQSTFGZ

ZEXAZAVVWO